

'Medical advancements have not created a better world for mankind.' Do you agree?

Key Words

'Medical advancements'

- The generation of information or a discovery of knowledge that improves our understanding in medical science, this is a significant breakthrough that results in an improvement of the diagnosis, treatment, and prevention of disease. Examples of medical advancements include the use of mRNA technology to create vaccines, the use of Artificial Intelligence (AI) for detecting of diseases, Immunotherapy for cancer treatments, CRISPR gene-editing technology, 3D printing to create implants and artificial prosthetics.

'Better world'

- Essays will need to consider how the advancements in medical science have brought about a change for humanity by making explicit comparisons to the past. Ideas of what constitutes a 'better' state for human society can vary, but they generally revolve around things like having greater equality and justice, more respect for the rights and dignity of the individual, reducing suffering, prolonging the lives of human beings, improve quality of life, etc. The scale and scope of impact should be broad and global in order to address 'better world'.

'For mankind'

- The range of examples discussed must address the benefits made available to the world at large. They should not, for example, discuss how individuals have benefitted from medical advancements.

Stand

While there are instances of medical advancements leading to challenges for mankind, they have on the whole achieved what they are generally purposed for: improving quality of life and longevity for humans.

Possible Approach

Arguments against the statement (why medical advancements have created a better world for mankind)

A1. Medical advancements have grown exponentially to become extremely effective in treating the causes and symptoms of a variety of human diseases/afflictions, either through more accurate diagnoses, or expanding the range of cures for various human illnesses that have plagued humankind throughout history. They have helped not only to considerably extend life expectancy but also to substantially reduce the extent/severity of human suffering, creating a better world for mankind.

- Over the past decade, for example, we have seen ground-breaking developments in HIV research, to the point that the AIDS epidemic that so fatally plagued much of the world population in the 1980s has declined in impact to become a chronic, manageable condition. This is largely thanks to the advances in medical care over the last 30 years, with the advent of antiretroviral therapy (ART) and once-daily single tablet regimens such as pre-exposure prophylactic pills (PrEP). Indeed, as of 2020, there has been a 31% decline in new HIV infections since 2010, and 75% of those infected with HIV globally receiving sustained antiretroviral therapy.

A2. Beyond just dealing with diseases and afflictions, medical advancements have also provided mankind with more freedom of choice, agency, and autonomy over various aspects of their lives, especially in the field of genetic modifications and enhancements, which have seen exponential growth over the past few years, and can increasingly provide humans with the capacity to reverse hitherto permanent ailments such as congenital/genetic diseases/disfigurations/deformities, etc.. leading to improved quality of life.

- Congenital anomalies are one of the leading causes of global disease, responsible for a staggering 57.7 million disability-adjusted life years lost worldwide (<https://www.ncbi.nlm.nih.gov/books/NBK333522/>). However, thanks to the advances made in medical technology, great strides have been made during the past 50 years in the diagnosis and management of said congenital anomalies. Formerly fatal or debilitating conditions can now be treated with success rates exceeding 90 percent, with the development of medical treatments such as patent ductus arteriosus ligations, atrial/ventricular septal defect repair, and total anomalous pulmonary venous return correction (<https://medlineplus.gov/ency/article/002948.htm>), with the result being that the patients consistently reported an increased quality of life.

Arguments in favour of the statement (why medical advancements have NOT necessarily created a better world for mankind)

B1. Medical advancements have, in fact, led to the creation of greater threats than before, with viral mutations and biological weapons becoming more of a present danger as medical advancements evolve. This is especially important given how some research facilities can deliberately tinker with or enhance viruses in order to test the efficacy of medicines, leading to greater risks of bio-leakages.

- As researchers continue to transition from the era of DNA sequencing into the era of DNA synthesis, it may soon become feasible to synthesise any virus whose DNA sequence is known. This was first demonstrated in 2001 when Dr. Eckard Wimmer re-created the poliovirus and again in 2005 when Dr. Jeffrey Taubenberger and Terrence Tumpey re-created the 1918 influenza virus. With the growing accessibility of DNA synthesis technology, computational power, and information, a growing number of people will have the capacity to produce bioweapons. For example, scientists have been able to transform the four letters of DNA (ACGT) into the ones and zeroes of binary code, making genetic engineering a simple matter of electronic manipulation. According to former American Secretary of State Hillary Clinton, 'the emerging gene synthesis industry is making genetic material more widely available [...] a crude but effective terrorist weapon can be made using a small sample of any number of widely available pathogens, inexpensive equipment, and college-level chemistry and biology' (<https://sites.dartmouth.edu/dujs/2013/03/10/genetically-engineered-bioweapons-a-new-breed-of-weapons-for-modern-warfare/>) .

B2. Medical advancements have surfaced ethical considerations and dilemmas that have hitherto not existed, with naysayers arguing that such scientific discoveries have led to the seeming decline of morality.

- With the creation of in-vitro fertilisation technology (IVF) came a whole slew of ethical controversies, suggesting that to some at least, medical advancements have not necessarily led to the betterment of humanity. Detractors of IVF have pointed out that preimplantation genetic diagnosis/screening processes, for example, raise problematic questions about the morality of seemingly eugenicist motivations, especially because individuals are now able to select for embryos based on histocompatibility for the donation of tissues to a sick family member, the diagnosis of genetic susceptibility to disease, and sex selection. In 2008, for example, a deaf British couple, Tom and Paula Lichy petitioned to create a deaf baby using IVF, leading to medical ethicists arguing that such a decision could potentially pave the way for designer babies and 'playing God' in the long run.

B3. While medical advancements made for the purpose of improving human life is indisputably a good thing, not everybody on the planet can have equal access to them. Indeed, it is often the case that only the minority of the rich and powerful get access to medical treatments routinely denied to the poor and underprivileged masses.

- While the advances made in the field of COVID-19 research have undoubtedly been effective and even led to the rapid development of COVID-19 vaccines, the benefits of such research are not distributed equally across all countries. As it is, the

statistics are sobering: by the end of November 2021, around 54.2% of the global population had received at least one dose of vaccine. For lower-income countries – the vast majority of which are in Africa –, the rate was just 5.8%. For as long as the inequitable distribution of COVID-19 vaccines remain, there will continue to be recurrent waves of COVID-19 infections, given that mutant COVID-19 outbreaks will persist so long as the entirety of the world population is unable to be completely vaccinated, and in time, reducing the efficacy of each dose. Indeed, where the disparity in vaccine allocation is higher, earlier and larger outbreaks of new COVID-19 waves will happen, limiting the benefits of COVID-19 prevention research and discoveries.